

Canning vegetables - demonstration lesson plan

A step-by-step guide for the volunteer leading the demo

Audience	Adults and teens (public outreach demonstration)
Group size	Any - works at a fair table or in a teaching kitchen
Time	About 60-75 minutes (pressure canning runs longer than water bath)
Format	Live demonstration with narration and questions
You should be	An Experienced Master Food Preserver volunteer

New to leading a demo? Read this first.

You do not have to memorize anything. This guide is written to be followed in order - it is completely fine to hold it and read from it during the demo.

Your job is to show the steps slowly, say why each one matters, and keep things safe. Bold black text tells you what to DO; lines in quotes are examples of what to SAY.

Walk through the steps once on your own before the event, and make sure you know how YOUR pressure canner vents and seals - dial vs weighted gauge differ. Lay equipment out in the order you will use it.

This demo is about a LOW-ACID food, so safety is the whole point. Never improvise a time, a pressure, or a shortcut.

If you are unsure, say so and look it up.

It is okay to not know an answer. Say "Great question - let me look that up in the OSU Extension MFP app so I give you the right answer," and move on. Never guess about safety.

Learning objectives

By the end of this demonstration, participants will be able to:

1. Explain why low-acid vegetables must be PRESSURE canned, not water-bath canned (botulism risk).
2. Set the correct canner pressure for their gauge type (dial or weighted) and altitude using the cited table.
3. Describe the pressure-canning sequence: vent 10 minutes, reach pressure before timing, hold pressure, then cool naturally.
4. Find the exact tested time for any vegetable in the official chart (PNW 172 / USDA) instead of guessing.

Agenda at a glance

The demo has five parts. Approximate timing - the processing run itself is long, so plan to narrate while the canner comes up to pressure:

- 1. Welcome & safety briefing (5 min)
- 2. Introduction: why vegetables need a pressure canner (7 min)
- 3. Live demonstration (the pressure-canning sequence) (30-40 min)
- 4. Questions & discussion (10 min)
- 5. Wrap-up & cleanup (5 min)

Before participants arrive - set up

do this first

Set up and inspect everything before anyone is watching. Know your canner. The full pack-and-inspect list is on the equipment checklist - bring these:

- Pressure canner with an accurate dial or weighted gauge
- Rack to keep jars off the bottom of the canner
- Mason jars suitable for canning
- Two-piece canning lids (flat lid and screw band)
- Jar lifter for safely handling hot jars
- Funnel to fill jars without spilling
- Bubble tool (plastic or rubber spatula) to remove air bubbles
- Timer to track processing times

Canning vegetables - demonstration lesson plan (cont.)

A step-by-step guide for the volunteer leading the demo

1. Welcome & safety briefing

read aloud, 5 min

Greet the group, introduce yourself as an OSU Master Food Preserver, and read this aloud before you touch any hot equipment:

- Low-acid vegetables must be processed in a pressure canner to destroy *Clostridium botulinum* spores that cause botulism poisoning.
- Boiling water bath canners do not reach the temperature needed (240 deg F) to safely process low-acid foods.
- Use only laboratory-tested processing times and pressures for safety.
- Improper processing risks botulism toxin formation and foodborne illness.

2. Introduction - why vegetables need a pressure canner

7 min

Hold up a jar of home-canned green beans and ask why we cannot just boil these in a water bath like jam or pickles. Let a few people answer, then explain:

Say something like:

"Vegetables are LOW-ACID foods. A boiling water bath only reaches 212 degrees - that is not hot enough to be safe. We have to use a pressure canner, which reaches about 240 degrees and destroys the spores of the bacterium that causes botulism."

"Botulism has no taste, no smell, and no look - you cannot tell by checking the jar. The only protection is processing at the right pressure for the full tested time."

"So two things are non-negotiable today: a pressure canner with an accurate gauge, and the exact tested time from the chart - never a water bath, and never a guessed or shortened time."

3. Live demonstration

go slow, narrate the WHY, 30-40 min

Work through each step out loud. The bold line is what to do; "Why" is what to tell them; the red line is a safety point to say directly. The pressure phase is the heart of this demo - slow down and explain venting, reaching pressure before timing, and natural cooldown.

1. Prepare vegetables by washing and cutting as desired.

Why it matters: Cleaning and cutting vegetables ensures even heat penetration and food safety.

Tell your audience: Always wash hands and produce thoroughly to reduce contamination.

2. Prepare jars and lids by washing in hot, soapy water; keep jars hot until filling to prevent breakage.

Why it matters: Hot jars prevent thermal shock and maintain food safety.

Tell your audience: Handle hot jars carefully to avoid burns.

3. Pack vegetables into jars using either a hot pack (precook vegetables before filling) or raw pack (fill raw vegetables directly).

Why it matters: Hot pack helps remove air and improve heat penetration while raw pack is quicker but may require longer processing.

Tell your audience: Follow tested recipes exactly; do not alter pack method or headspace.

4. Leave 1 inch of headspace between the top of the food and the jar rim.

Why it matters: Proper headspace allows for food expansion and forms a good vacuum seal.

Tell your audience: Too little headspace can cause food to expand and leak; too much can cause poor seals.

5. Remove air bubbles by gently sliding a bubble remover or plastic spatula around inside the jar.

Why it matters: Air bubbles can interfere with heat penetration and sealing.

Tell your audience: Be careful not to scratch jars.

6. Wipe jar rims with a clean, damp cloth to ensure a clean sealing surface.

Why it matters: Clean rims are essential for proper lid seal.

Tell your audience: Do not reuse lids to ensure a safe seal.

Canning vegetables - demonstration lesson plan (cont.)

A step-by-step guide for the volunteer leading the demo

7. Apply the flat lid and screw the metal band down fingertip tight.

Why it matters: Proper lid application prevents leakage while allowing air to escape during processing.

Tell your audience: Do not overtighten screw bands before processing to avoid lid damage.

8. Place jars on the rack in the pressure canner filled with 2-3 inches of simmering water.

Why it matters: Water protects jars and helps transmit heat evenly.

Tell your audience: Avoid pouring water directly on jars when filling the canner.

9. Lock the canner lid securely, following manufacturer instructions.

Why it matters: A secure lid ensures the canner builds and maintains the necessary pressure.

Tell your audience: Improper lid placement can cause dangerous steam leaks.

10. Vent or exhaust steam from the canner for 10 minutes by opening the petcock or vent.

Why it matters: Removing air ensures that the canner reaches 240 deg F uniformly.

Tell your audience: Keep hands and face away from escaping steam to avoid burns.

11. Close the vent or petcock and bring the canner to the correct pressure for your altitude and gauge type.

Why it matters: Correct pressure is essential to reach the temperature required to kill botulism spores.

Tell your audience: Watch pressure gauge carefully to avoid under or over-pressurizing.

12. Start timing the process only after the correct pressure is reached and maintain this pressure for the full processing time.

Why it matters: Accurate timing at correct pressure ensures safety and food quality.

Tell your audience: Do not open the canner or reduce pressure during processing.

13. When processing time ends, turn off heat and let the pressure return to zero naturally-do NOT force-cool or open the canner early.

Why it matters: Slow depressurizing prevents liquid loss and jar breakage.

Tell your audience: Never attempt to speed cooling by removing the weight or running cold water on the canner.

14. After pressure is fully released, wait 10 minutes before carefully removing the lid, tilting the lid away from you.

Why it matters: This waiting time reduces risk of steam burns.

Tell your audience: Always remove lid away from face to avoid steam injury.

15. Remove jars using jar lifter and place on a towel or rack, spaced apart to cool for 12-24 hours undisturbed.

Why it matters: Cooling allows jars to seal and prevents spoilage.

Tell your audience: Do not retighten metal bands while lids are hot.

16. After cooling, check seals by pressing the center of each lid; a sealed lid will not flex up or down.

Why it matters: Ensures food safety by confirming vacuum seal.

Tell your audience: Discard or refrigerate unsealed jars to prevent foodborne illness.

4. Questions & discussion

OSU-sourced answers, 10 min

Invite questions: "What questions do you have so far?" works better than "Any questions?" Below are the ones you will hear most, with answers you can trust. If you get a question not listed here, it is fine to look it up in the MFP app rather than guess.

Q: Why must vegetables be pressure canned instead of using a boiling water bath?

A: Low-acid foods such as vegetables must be processed in a pressure canner at 240 deg F to destroy botulism spores and other spoilage organisms. Boiling water bath, which reaches only 212 deg F, is not sufficient for these low-acid foods. Pressure canning ensures safe preservation by reaching the required temperature to prevent spoilage and botulism.

Source: SP 50 445 Preserving green italian or wax beans

Canning vegetables - demonstration lesson plan (cont.)

A step-by-step guide for the volunteer leading the demo

Q: How do I vent (exhaust) the canner before pressurizing?

A: To vent a pressure canner, steam must be allowed to escape steadily from the petcock for 10 minutes before closing the petcock or placing the weighted gauge on the canner. This venting removes all the air from the canner to ensure adequate heating of jars. This step is required even for canners labeled 'self-venting' or 'self-exhausting'.

Source: PNW 172 Canning vegetables

Q: When do I start counting the processing time?

A: Start timing the process when the pressure reading on the dial gauge indicates the recommended pressure has been reached, or when the weighted gauge begins to jiggle or rock as described by the manufacturer.

Source: PNW 300 Canning tomatoes and tomato products

Q: What do I do if the pressure drops below the required level during processing?

A: If the pressure drops below the recommended level at any time during processing, bring the canner back to pressure by increasing the heat, and begin the timing process over again using the original processing time.

Source: PNW 300 Canning tomatoes and tomato products

Q: How do I cool the canner and remove jars safely?

A: When processing time is complete, turn off the heat and remove the canner from the heat source. Allow the canner to cool undisturbed until it is fully depressurized. Then remove the weight or slowly open the petcock, wait 10 minutes, and carefully open the lid away from you to avoid steam burns. Use a jar lifter to remove jars, keeping them upright, and place on a heavy towel or cooling rack with space between jars to cool undisturbed.

Source: PNW 300 Canning tomatoes and tomato products

5. Wrap-up & cleanup

5 min

Close by reinforcing the main takeaways, then clean up:

- Recap what makes canned vegetables safe: a pressure canner with an accurate gauge, the correct pressure for your altitude, and the full tested time from the chart.
- Remind them low-acid vegetables are NEVER safe in a boiling water bath.
- Point them to help: "For the exact time for any vegetable, use PNW 172 or the USDA guide. When in doubt, check the OSU Extension MFP app or call the Extension office - never guess."

Then clean up:

- Turn off heat and let the canner cool fully before opening or moving it.
- Wash and dry all equipment; store jars in a cool, dark place.
- Label sealed jars with contents and date; refrigerate or reprocess any unsealed jars within 24 hours.

Sources

Every value above is drawn from these OSU/USDA Extension publications. Verify against the cited source before public use.

- PNW 172 Canning vegetables
- PNW 300 Canning tomatoes and tomato products
- SP 50 305 Preserving peas edible pea pods
- SP 50 443 Preserving corn
- SP 50 445 Preserving green italian or wax beans